

Project Design Phase-II

Solution Requirements (Functional & Non-functional)

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Project Name	Automated Network Request Management in ServiceNow

Functional Requirements

Following are the functional requirements implemented across the ServiceNow Service Portal form (/sp), Flow Designer, and custom Database Tables schema.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	Service Portal Catalog Intake	• Access 'Network Request' item under Network request Management on /sp • Auto-populate Requested For with current logged-in user • Support Opened on behalf of delegation lookup • Capture User Name, Phone Number, Email ID, and Mobile Number
FR-2	Connection & Billing Configuration	• Select mandatory Type of Connection (New or Existing) • Calculate and enter Total Amount for the connection service • Select Mode of Payment via radio choices (UPI or Card) • Enter multi-line delivery Address and attach supporting documents
FR-3	Dynamic Form Display & Validation	• Enforce mandatory fields to prevent empty or incomplete submissions • Implement Catalog UI Policies for dynamic field behavior • Validate email and phone formatting prior to

		form submission
FR-4	Custom Table Record Persistence	• Flow Designer Step 1 extracts all submitted catalog variables • Flow Designer Step 2 creates a new record in Database Tables (u_database_table) • Generate unique Auto-Numbered ticket identifier for request tracking • Map all portal inputs (user, connection, payment, address) to table fields
FR-5	Automated Requester Acknowledgment	• Flow Designer Step 3 dispatches automated acknowledgment email to Requester • Include record auto-number, submission summary, and initial status
FR-6	Approval Routing & Governance	• Flow Designer Step 4 initiates 'Ask For Approval' on the created Database Tables record • Route approval requests to designated Manager / Admin • Track approval decisions directly in sysapproval_approver related records
FR-7	Approval State Branching & Execution	• Flow Designer Step 5 evaluates 'If Request is Approved' condition • True Branch (Step 6 & 7): Update Database Tables record status to Approved and send approval confirmation email to requester • False Branch (Step 9 & 10): Update Database Tables record status to Rejected and send rejection notification email to requester

Non-functional Requirements

Following are the non-functional requirements governing the application's performance, security, and maintainability.

NFR No.	Non-Functional Requirement	Description
NFR-1	Usability	The Service Portal interface (/sp) must be clean and intuitive. Requesters should be able to complete a Network Request within 2 minutes with auto-populated profiles and guided dynamic fields.
NFR-2	Security	Enforce role-based Access Control Lists (ACLs) on Database Tables (u_database_table). Requesters can only view their own submissions, while approvers and fulfillers have managed write/approval permissions.
NFR-3	Reliability	100% of submitted requests must trigger Flow Designer without lost transactions. All approval states and state changes must be persistently logged for audit traceability.
NFR-4	Performance	Catalog page load time on the Service Portal must remain under 2 seconds. Flow Designer actions (record creation, email triggers, and approval requests) must execute within 3 seconds of form submission.
NFR-5	Availability	The Network Request portal service must be available 24/7 on the ServiceNow cloud instance, allowing users to submit requests at any time without downtime.
NFR-6	Scalability	The system architecture must support concurrent user submissions and handle growing ticket volumes in u_database_table without manual table re-indexing or performance drops.